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1. Introduction

Optical coherence tomography angiography (OCTA) is an extension of OCT that visualizes vascular perfusion using the motion contrast from red blood cells (RBCs). Over the years, OCTA has demonstrated widespread acceptance and use for clinical evaluations of vascular features in relation to disease [1]. OCT angiograms can be used to evaluate characteristics on the scale of vascular layers by measuring vascular density or non-perfusion zones, or it can be used to examine individual vessels with measurements of diameter or tortuosity. Nevertheless, OCT is prone to certain artifacts which specifically affect visualization of the vasculature. These artifacts may lead to underestimations of measurements of vessel diameter and density. When viewed in cross-section, large vessels often demonstrate an hourglass appearance with stronger scattering in the center than at the sides (Figure 1A). This appearance is due to orientation effects of RBCs, which align with the blood flow, causing a reduced scattering cross-section at the sides of the vessels. To compensate for these artifacts, OCT contrast agents such as Intralipid have previously been used to increase intravascular intensity signals, particularly in regions where RBC scattering is diminished. Here we examine different methods of measuring vessel size from in vivo OCT scans of mouse retinal vasculature, and we use a contrast agent to obtain ground truth vessel diameter measurements for comparison. We finally propose ways of mitigating or correcting for underestimations of vessel diameter related to typical OCT artifacts.

Optical coherence tomography (OCT) is a non-invasive optical imaging method, which uses interferometry to measure the transit time of light that is backscattered from a sample. OCT has found widespread clinical use in ophthalmology with XXX number of systems and XXX OCT scans performed annually around the world. OCT angiography (OCTA) is a functional extension of OCT which highlights motion between repeated OCT scans to generate images of vascular networks. Typically, OCTA signals are first calculated from the OCT intensity and/or phase information, segmented in 3D manually, algorithmically, or using a deep learning approach, and finally compressed into a 2D image using a maximum intensity projection. These 2D vascular maps can then be binarised by using a global or adaptive threshold, and finally vascular metrics such as vessel density and diameter can be computed. Of these processing stages, the thresholding step is perhaps the most important. There are dozens of different thresholding methods that have been proposed over the years and each of these may produce different results in the binary map, which would in turn influence the quantification of vascular metrics. In this work, we examine an alternate approach for extracting quantitative vessel diameter by fitting a model to the OCTA data before binarisation to avoid thresholding-based effects.

In addition to thresholding-based variability, we hypothesize that OCTA metrics are also affected by OCT artifacts caused by the non-uniform backscattering profiles of red blood cells (RBCs). Because RBCs have a distinct bi-concave disc shape, they more effectively backscatter light when aligned perpendicular to the beam, rather than parallel to it. It has already been observed that in large vessels, a so-called hourglass artifact appears when viewed cross-sectionally. This is caused by the alignment of the RBCs with the vessel walls while under laminar flow, where the RBCs are perpendicular to the beam in the center of the vessels and parallel to the beam at the edges of the vessels. This artifact artificially suppresses the OCT and OCTA signals at the edges of the vessels, making them appear more narrow when projected into an en-face view, which is the preferred way of displaying angiographic data. Here we use contrast enhanced OCT (CE-OCT) to investigate how significant these effects are and how they can be corrected.

2. Methods and Materials

2.1. Imaging system

All imaging was performed using a custom-built polarisation-sensitive spectral-domain OCT ophthalmoscope, described in detail previously [2]. The system was centred at 840 nm with a spectral bandwidth of 100 nm at full width half maximum (FWHM), providing an axial resolution of approximately 5.1 µm in air and 3.8 µm in tissue (assuming a refractive index of 1.35). A sensitivity of 96 dB was achieved at an A-scan rate of 80 kHz and a beam diameter of 0.5 mm at the pupil plane (power at cornea = 2.85 mW) [3]. This system was employed to acquire volumetric OCT data from mouse retinas before, during, and after an intravascular contrast agent administration, as well as before and after functional stimulation of the mouse retina using a custom 502 nm light-emitting diode (LED) ring flashing at 10 Hz (power at cornea = 26.6 μW). This wavelength was selected to target the peak absorption of mouse rhodopsin (~500 nm), enabling predominantly rod-driven photoreceptor stimulation [4].

2.2. Experimental protocol

Three mouse models were used in this study: VLDLR knockout mice (B6;129S7-*Vldlr*^{tm1Her}/J, The Jackson Laboratory, Bar Harbor, USA), C57BL/6 mice (C57BL/6, The Jackson Laboratory, Bar Harbor, USA), and SOD1 transgenic mice (strain details, The Jackson Laboratory, Bar Harbor, USA), with n = XX mice per group. All imaging sessions were conducted under general anaesthesia using either isoflurane or a ketamine/xylazine combination. Isoflurane was induced at 4% in oxygen for 4 minutes and subsequently maintained at 2% in oxygen at a flow rate of 1.5–2 L/min. In cases where isoflurane alone did not provide adequate sedation, an intraperitoneal injection of ketamine/xylazine cocktail (100 mg/kg ketamine, 6 mg/kg xylazine; 10 mL/kg body weight) was administered. To maintain core body temperature of the mice throughout imaging, they were positioned on a heating pad. Mouse pupils were dilated with one drop of tropicamide (5 mg/mL, Agepha Pharma s.r.o., Senec, Slovakia) per eye, and artificial tear drops (Oculotect, ALCON Pharma GmbH, Freiburg im Breisgau, Germany) were applied at regular intervals to prevent the cornea from drying out.

Intralipid has been shown to be a highly scattering contrast agent that fills the vascular lumen uniformly, thereby compensating for the artefact caused by the diminished backscattering at vessel peripheries caused by RBC orientation effects [3, 5, 6]. For this study, Intralipid 20% (3 mL/kg body weight) was administered as a bolus injection via the tail vein. Volumetric angiography data were acquired over a 1 mm × 1 mm field of view centred on the optic nerve head (ONH), with a scan density of 500 A-scans per B-scan and five repeated B-scans at each of 400 slow-axis positions (acquisition time approximately 16 seconds). For this study, pre- and post-injection datasets and pre- and post-stimulation datasets were acquired. Additionally, a dynamic-contrast OCT (DyC-OCT) protocol [7] was employed during the contrast injection or during stimulation, consisting of 2000 repeated B-scans over approximately 16 seconds (interscan time 8.2 ms) at the same cross-sectional location over a 1 mm lateral field of view. DyC-OCT scans allow us to track the same vessels over time to precisely measure the effects of the contrast agent or the stimulation with a high degree of spatial coregistration. All animal procedures were approved by the ethics committee of the Medical University of Vienna and the Austrian Federal Ministry of Education, Science, and Research (BMBWF/66.009/0272-V/3b/2019 and GZ 2024-0.802.524).

2.3. Post-processing

Cross-polarised OCT intensity data from the retinal pigment epithelium (RPE) were used for inter-frame alignment and B-scan flattening [3, 8]. Angiograms were derived by applying the complex subtraction OCT angiography (OCTA) algorithm [9, 10]. To enhance vessel visibility and suppress background noise, a 3D averaging filter with a kernel size of 50 × 50 × 10 voxels was applied to both the angiogram and DyC-OCT volumes. For the volumetric angiogram data, the kernel dimensions correspond to the lateral fast-axis (x), lateral slow-axis (y), and axial (z) dimensions, while for the DyC-OCT data, the kernel spans the lateral (x), axial (z), and temporal (t) dimensions. All data processing and analysis was performed in MATLAB (R2019b, MathWorks, Natick, MA, USA).

2.4. Diameter measurement

We evaluated multiple combinations of methods for quantifying vessel diameter from the OCTA data to determine which combinations of steps work best under different conditions. Specifically, we looked at two types of data projections (mean intensity and maximum intensity) to generate vessel profiles and several different quantifications of those profiles including a traditional binarisation, a Gaussian model approach, and a Generalised-Gaussian (GG) model approach. These different methods were evaluated against the ground truth as described in this section.

2.4.1. Ground truth calculation

Ground truth vessel diameters were obtained from contrast-enhanced B-scan angiograms, where the uniformly scattering Intralipid fills the full vascular lumen regardless of RBC orientation. The width and the height of the vessel were manually measured from the vessel's cross-section. The ground truth diameter was defined as the minimum of the two measurements to account for off-axis cross-sectioning of the vessel, which produces a more elliptical appearance in the B-scan. For cases where the vessel was perpendicular to the B-scan, the height and width were in agreement. Because the lateral diameter is studied here, we defined a correction factor of $height/width$ for cases where the width was larger than the height due to skew effects and was equal to 1 otherwise. By multiplying this correction factor against diameter measured from the ROI projection, we ensured that the measured vessel diameter reflected the true size of the vessel.

2.4.2. Vessel projection calculation

For each vessel of interest, a rectangular region of interest (ROI) encompassing the vessel cross-section, along the lateral (x) and axial (z) dimensions, was selected from the DyC-OCT B-scan angiograms. En-face intensity profiles were then generated by projecting the A-scans within this ROI along the axial (z) direction yielding a time-stack of 1-D lateral intensity profiles for each vessel. Two projection methods were compared: maximum intensity projection (MIP), which selects the highest intensity value along the depth axis of each ROI, and mean intensity projection (MeIP), which averages the values along each ROI.

2.4.3. Vessel diameter calculation

Two parametric models were then fitted (Fig. 1a) to these one-dimensional intensity profiles from the previous step, using nonlinear least-squares regression in MATLAB.

The first model was a standard Gaussian function:

$$f(x) = A \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) + C \tag{1}$$

where A is the peak amplitude, μ is the centre position, σ is the standard deviation, and C is a constant baseline offset.

The second model was a Generalised-Gaussian (GG) function:

$$f(x) = A \exp\left(-\left(\frac{|x-\mu|}{\alpha}\right)^{\beta}\right) + C \tag{2}$$

where A is the peak amplitude, μ is the centre position, α is the scale parameter, β is the shape parameter, and C is a constant baseline offset. Given the imaging system configuration, all vessels are expected to produce at minimum a Gaussian intensity profile in the B-scan cross-section, and so profiles sharper than Gaussian profile ($\beta < 2$) are therefore not technically meaningful. The shape parameter β was hence constrained to $\beta \geq 2$ during fitting. With this constraint, the GG accommodated to profiles ranging from Gaussian ($\beta = 2$, where $\sigma = \alpha/\sqrt{2}$) to increasingly flat-topped ($\beta > 2$).

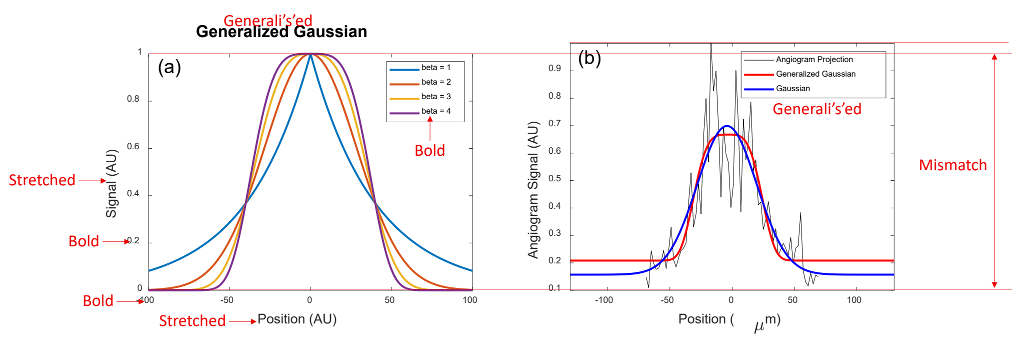

Fig. 1. (a) GG function for shape parameters $\beta = 1, 2, 3$, and 4. At $\beta = 2$ the function reduces to a standard Gaussian; higher β values produce increasingly flat-topped profiles. (b) Example en-face vessel intensity profile (black) with standard Gaussian (blue) and GG (red) fits.

Two standard width metrics were calculated from the fitted models: the full width at half maximum (FWHM) and the $1/e^2$ width. For the standard Gaussian fit, the $1/e^2$ width was calculated as:

$$w_{1/e^2} = 4\sigma \tag{3}$$

The FWHM for a Gaussian is related to the $1/e^2$ width by a fixed proportionality:

$$\text{FWHM} = 2\sigma\sqrt{2\ln 2} = \frac{\sqrt{2\ln 2}}{2}\, w_{1/e^2} \tag{4}$$

and therefore only the $1/e^2$ width was calculated directly from the fit, with the FWHM obtainable via this known relationship.

For the GG fit, no such linear relationship exists between the FWHM and $1/e^2$ width, since both depend on the shape parameter β. Both metrics were therefore calculated independently as:

$$\mathrm{FWHM}_{\mathrm{GG}} = 2\alpha(\ln 2)^{1/\beta} \tag{5}$$

$$w_{1/e^2,\,\mathrm{GG}} = 2\alpha \cdot 2^{1/\beta} \tag{6}$$

For the binary mask approach, multiple threshold values were chosen for testing, and the resulting vessel diameters were compared to the ground truth. For each C-scan, the maximum angiogram intensity within the smoothed volume was calculated, and the threshold values were selected as a fixed percentage of this maximum intensity. The vessel diameter was then measured as the number of pixels above each of the threshold values across the vessel cross-section.

3. Results

3.1. Qualitative assessment of vessel underestimation

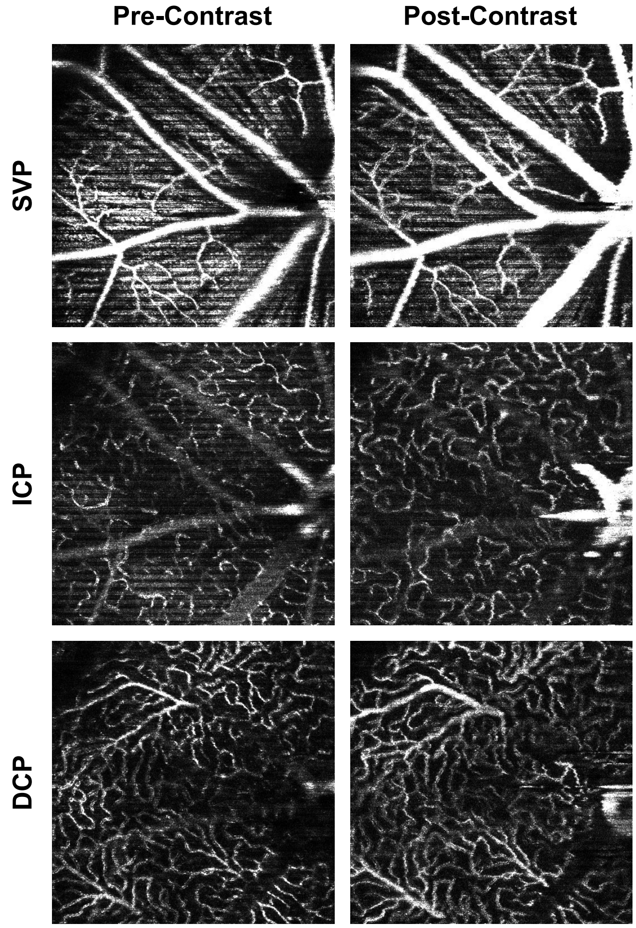

Fig. 2. OCTA MIP en face of the SVP, ICP, and DCP before (left) and after (right) the administration of Intralipid 20%. Vessels appear visibly wider and brighter post-contrast across all layers, most prominently in the larger vessels.

Figure 2 shows OCTA MIP en face of the superficial vascular plexus (SVP), intermediate capillary plexus (ICP), and deep capillary plexus (DCP) before and after contrast injection. Across all three layers, vessels appeared visibly wider and brighter post-contrast injection, consistent with previous literature showing that OCTA underestimates vessel calibre in the absence of a contrast agent [3]. This effect was most pronounced in the larger vessels, especially in the SVP, where the major retinal arteries and veins appeared noticeably thicker after contrast administration. In the ICP and DCP, the capillary networks also showed improved visibility, though the effect was less dramatic.

DyC-OCT scans were acquired to visualise the dynamics of OCTA vessel diameter pre- and post-contrast injection (Fig. 3). Pre-contrast injection, the vessel cross-sections appeared narrower than their true diameter due lateral edge signal suppression by the RBC orientation artefact (Fig. 3a). At peak contrast, the Intralipid filled the vascular lumen uniformly and showed the full extent of the vessel cross-section (Fig. 3b). However, the high scattering nature of the contrast agent produced light diffusion beyond the vessel wall, resulting in an apparent vessel diameter that slightly exceeded the true anatomical diameter. In the post-contrast phase, the signal stabilised as the initial bolus cleared, and the vessel boundaries remained more visible than before injection without the overestimation observed at peak contrast (Fig. 3c). The MIP and the corresponding binary mask over time (Figs. 3d–e) illustrate the observed temporal changes in vessel diameter across the three phases. The OCTA mean intensity (Fig. 3f) showed a sharp increase post-contrast injection, followed by a decrease to a new steady state that remained elevated above the pre-contrast baseline. These observations confirm that the contrast agent does significantly enhance vessel visibility and diameter, and that the post-contrast images show improved vessel delineation compared to pre-contrast.

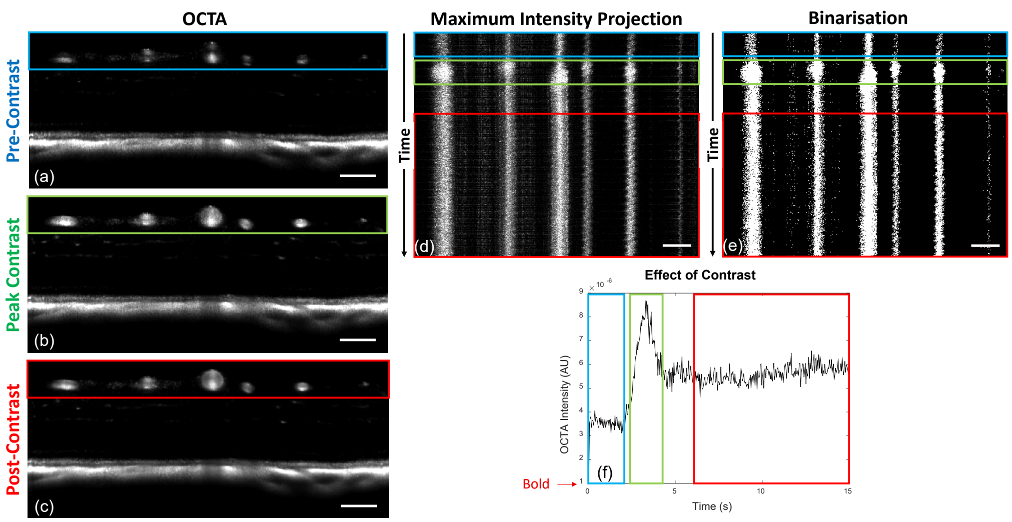

Fig. 3. DyC-OCT B-scan angiograms at (a) pre-contrast, (b) peak contrast, and (c) post-contrast timepoints. (d) MIP and (e) binarisation of the vessel cross-sections over time, with coloured boxes indicating the pre-contrast (blue), peak-contrast (green), and post-contrast (red) windows. (f) Mean OCTA intensity over time, showing the contrast bolus dynamics. All scale bars are 100 µm.

3.2. Linearity and accuracy of model-based methods

To evaluate the linearity and accuracy of the model-based measurements, the calculated diameter values were plotted against the ground truth calculations for each vessel across three contrast conditions: pre-contrast, peak contrast, and post-contrast (Fig. 4). Six combinations were evaluated: two projection methods (MIP and MeIP) by three model-width metrics (Gaussian $1/e^2$, GG FWHM, and GG $1/e^2$). A slope of $m = 1$ (black line) indicates perfect agreement with the ground truth, and the coefficient of determination (R^2) quantifies the goodness of fit of the linear model.

For the Gaussian $1/e^2$ width, pre-contrast measurements yielded slopes of $m = 0.70$ ($R^2 = 0.83$) for MIP and $m = 0.73$ ($R^2 = 0.85$) for MeIP, indicating substantial underestimation (Figs. 4a,b). With contrast enhancement, the slope values increased where peak contrast MeIP measurements showed $m = 1.04$ ($R^2 = 0.95$), and post-contrast showed $m = 1.04$ ($R^2 = 0.97$), which is close to the ideal slope of 1.

The GG FWHM metric consistently returned diameters well below the ground truth, with pre-contrast slopes of $m = 0.42$ (MIP) and $m = 0.43$ (MeIP), peak contrast slopes of $m = 0.66$ (MIP) and $m = 0.61$ (MeIP), and post-contrast slopes of $m = 0.66$ (MIP) and $m = 0.61$ (MeIP) (Figs. 4c,d). The GG $1/e^2$ width provided the best overall performance (Figs. 4e,f). MIP based GG $1/e^2$ measurements achieved close to unity at peak contrast ($m = 0.98$, $R^2 = 0.98$) and post-contrast ($m = 1.02$, $R^2 = 0.93$). MeIP based GG $1/e^2$ measurements showed strong agreement with the ground truth as well, with $m = 0.98$, $R^2 = 0.97$ at peak contrast and $m = 1.00$, $R^2 = 0.97$ at post-contrast. Pre-contrast MeIP GG $1/e^2$ measurements showed a slope of $m = 0.71$ ($R^2 = 0.88$), which was the highest pre-contrast R^2 among all the pre-contrast model evaluated width metrics.

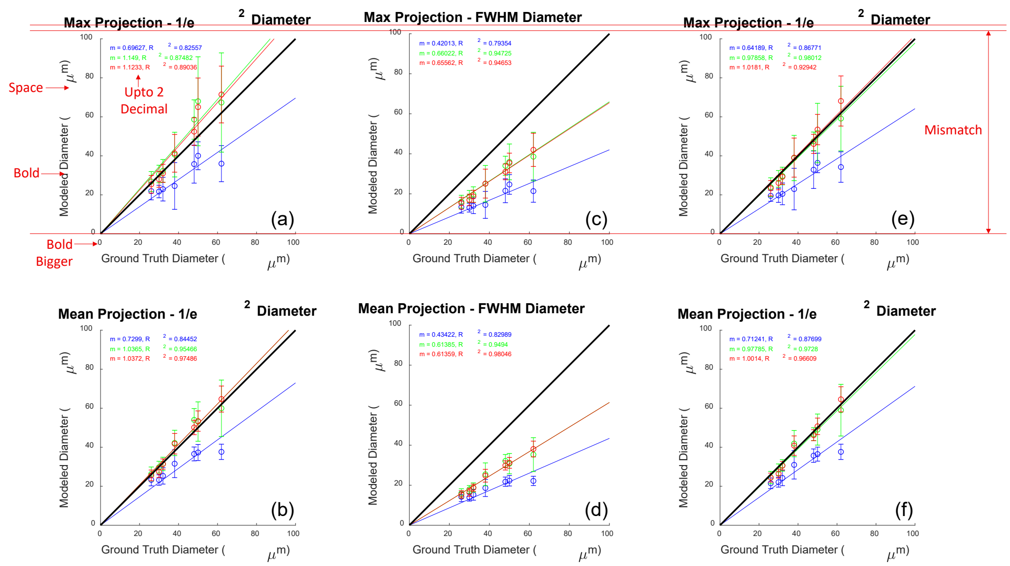

Fig. 4. Modelled versus ground truth vessel diameter using MIP (top row) and MeIP (bottom row). Columns correspond to (a,b) Gaussian $1/e^2$ width, (c,d) GG FWHM, and (e,f) GG $1/e^2$ width. Regression lines and statistics are shown for pre-contrast (blue), peak contrast (green), and post-contrast (red) conditions. The black line indicates the identity line ($m = 1$).

3.3. Linearity and accuracy of binarisation

Linearity and accuracy analysis was also performed for diameter measured from binarised black and white (BW) en face images at five threshold levels (BW-1 through BW-5, in order of ascending threshold values) for both MIP (Fig. 5) and MeIP (Fig. 6).

For MIP (Fig. 5), pre-contrast slopes values decreased with increasing threshold, from $m = 0.61$ ($R^2 = 0.58$) at BW-1 to $m = 0.24$ ($R^2 = 0.24$) at BW-5. Both slope and R^2 degraded substantially at higher threshold values. Peak contrast measurements showed a similar decline, from $m = 0.96$ ($R^2 = 0.88$) at BW-1 to $m = 0.65$ ($R^2 = 0.74$) at BW-5. Post-contrast measurements also showed a decline, from $m = 0.92$ ($R^2 = 0.76$) at BW-1 to $m = 0.54$ ($R^2 = 0.34$) at BW-5.

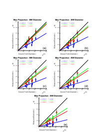

Fig. 5. Vessel diameter measured from BW images versus ground truth for MIP at five threshold levels: (a) BW-1 (lowest) through (e) BW-5 (highest). Regression lines and statistics are shown for pre-contrast (blue), peak contrast (green), and post-contrast (red). Both slope m and R^2 decrease with increasing threshold values.

For MeIP (Fig. 6), the pre-contrast slope decreased from $m = 0.83$ ($R^2 = 0.59$) at BW-1 to

$m = 0.53$ ($R^2 = 0.65$) at BW-5. Notably, the R^2 values remained relatively stable or slightly improved with increasing threshold, which was in contrast to the consistent decline observed for MIP. Peak contrast MeIP measurements at the intermediate threshold BW-3 showed $m = 0.99$ ($R^2 = 0.86$). However, at the lowest threshold BW-1, it slightly overestimated the ground truth ($m = 1.14$). Overall, MeIP provided more reliable binarisation-based vessel diameter measurements than MIP.

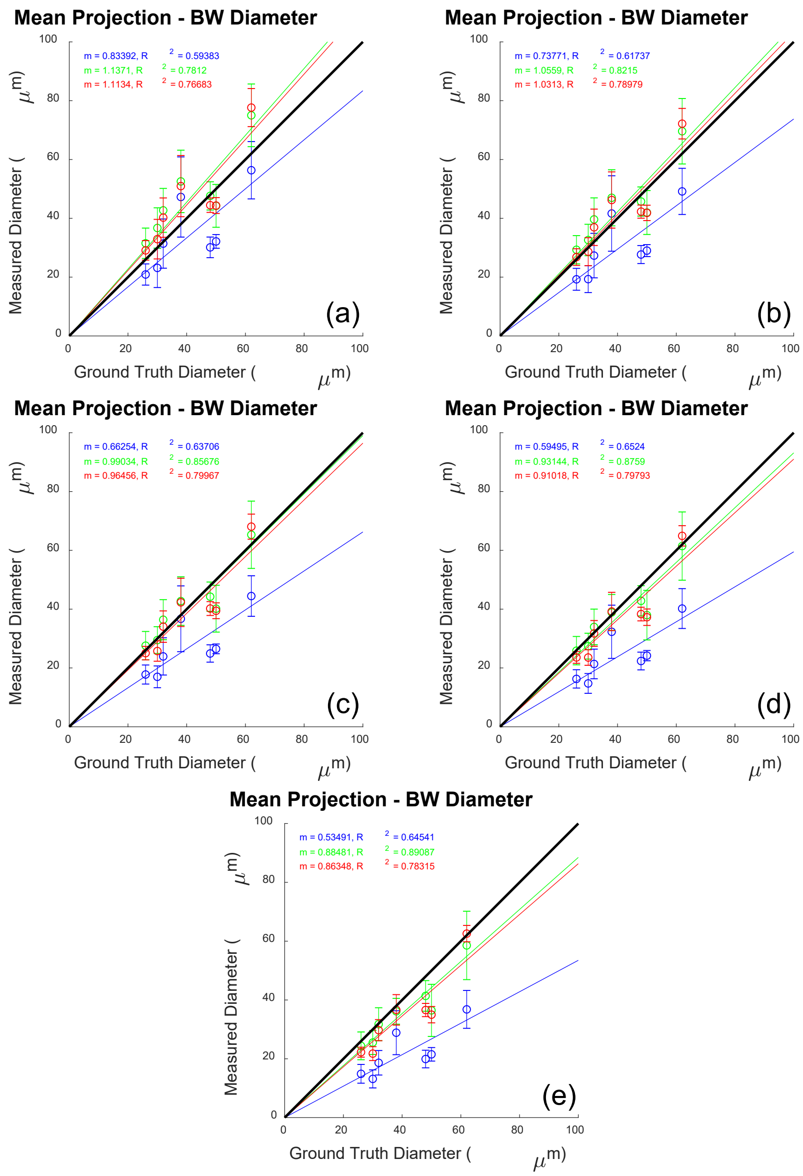

Fig. 6. Vessel diameter measured from BW images versus ground truth for MeIP at five threshold levels: (a) BW-1 (lowest) through (e) BW-5 (highest). Regression lines and statistics are shown for pre-contrast (blue), peak contrast (green), and post-contrast (red) conditions. MeIP yielded higher slopes and more stable R^2 values compared to MIP (Fig. 5).

3.4. Repeatability and linearity comparision across all methods

(confirm the values reported here with the actual data, as these are just extracted from the bar plots.)

To compare the repeatability and linearity across BW1–5 and the model based approach, the coefficient of variation (CoV) and R^2 were computed (Fig. 7). For MIP based measurements (Fig. 7a), the CoV of BW method increased with increase in threshold while the CoV of the model based methods remained relatively stable. With increase in threshold values BW1–5, the CoV increased from approximately 0.2 to 0.47 for pre-contrast, 0.17 to 0.32 for peak contrast, and 0.1 to 0.18 for post-contrast data. The model based methods (G-e^2, GG-HM, GG-e^2) exhibited CoV values of approximately 0.24–0.27 for pre-contrast data, with reduced values of approximately 0.16–0.18 post-contrast. For MeIP based measurements (Fig. 7b), the BW CoV values were lower and more stable across thresholds, with values observed as approximately 0.18–0.20 pre-contrast, 0.16–0.17 peak contrast, and 0.09–0.13 post-contrast. The model based CoV values were lower as well, with values observed as approximately 0.13–0.15 pre-contrast, increasing slightly to 0.13–0.16, and then decreasing to approximately 0.09–0.10 post-contrast.

The linearity results followed a complementary pattern. For MIP (Fig. 7c), the BW R^2 values declined sharply from approximately 0.59 at BW-1 to 0.23 at BW-5 for pre-contrast data, while the model based R^2 values ranged from 0.80 to 0.86 pre-contrast and reached 0.90–0.97 at peak and post-contrast. For MeIP (Fig. 7d), the BW R^2 values were more stable (approximately 0.60–0.65 pre-contrast) but still substantially lower than the model based values (0.83–0.85 pre-contrast, reaching 0.95–0.98 post-contrast). Taken together, these results demonstrate that model based methods consistently outperformed binarisation in both repeatability and linearity across all contrast conditions, and that MeIP based processing yielded superior performance compared with MIP for both method categories.

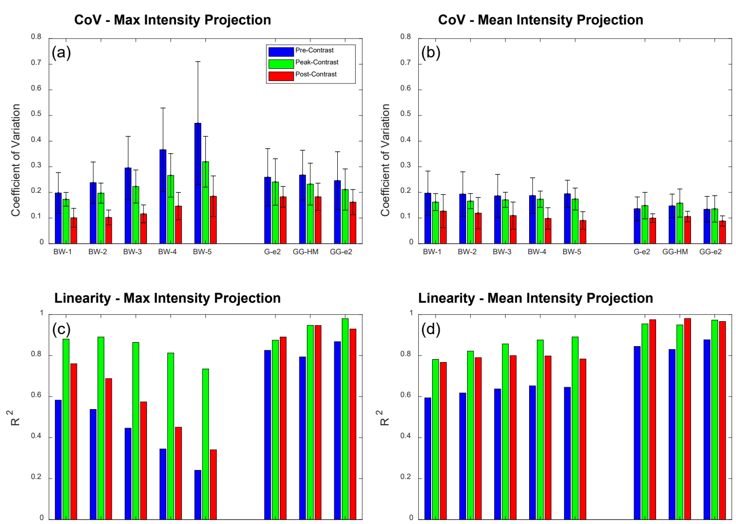

Fig. 7. Summary of repeatability and linearity across all diameter measurement methods. (a,b) Coefficient of variation (CoV) for MIP and MeIP, respectively. (c,d) Coefficient of determination (R^2) for MIP and MeIP, respectively. Bars are grouped by method (BW-1 through BW-5, G-e^2, GG-HM, GG-e^2) and colour-coded by contrast condition: pre-contrast (blue), peak-contrast (green), and post-contrast (red). Model based methods consistently show lower CoV and higher R^2 than binarisation based methods.

3.5. Functional flicker stimulation

To investigate whether the choice of diameter measurement method affects the detection of physiological vascular responses, functional flicker stimulation was performed in a C57BL/6 mouse and vessel diameter was measured before and after photoreceptor stimulation using each of the three approaches (Fig. 8).

The Gaussian and GG model-based time series (Figs. 8a,b) showed visible diameter increases following stimulation onset for most of the five vessels examined, consistent with the expected neurovascular coupling-driven vasodilation. The binarised angiogram time series (Fig. 8c), however, showed the opposite trend: a paradoxical decrease in measured diameter following stimulation. Bar charts summarising the mean pre- and post-stimulation diameters per vessel (Figs. 8d–f) confirmed this pattern. For the Gaussian model, four of five vessels showed significant diameter increases ($p < 0.01$), with percentage changes of +9.2%, +9.6%, +28.5%, and +4.9% for vessels 1, 2, 4, and 5, respectively (Fig. 1). The GG model similarly detected significant increases in three of five vessels (+9.9%, +28.4%, and +3.1% for vessels 2, 4, and 5). In marked contrast, the binarised method detected significant diameter *decreases* in four of five vessels (−12.5%, −10.2%, −4.2%, and −9.3% for vessels 1, 2, 3, and 5).

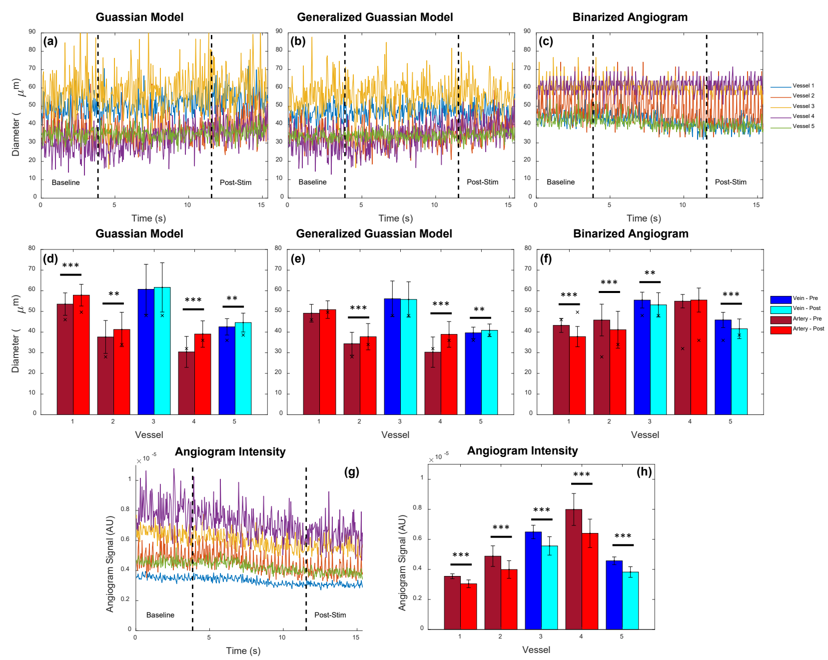

Fig. 8. Functional flicker stimulation results. (a–c) Time series of vessel diameter during baseline and post-stimulation periods for (a) Gaussian model, (b) generalised Gaussian model, and (c) binarised angiogram, across five vessels. Dashed lines indicate stimulation onset and offset. (d–f) Mean diameter per vessel before (dark bars) and after (light bars) stimulation for each method; veins are shown in blue/cyan and arteries in dark/light red. Significance levels: $^{**}p < 0.01$, $^{***}p < 0.001$. (g) Angiogram signal intensity time series and (h) mean intensity per vessel before and after stimulation. Model-based methods detect the expected vasodilation, whereas the binarised method paradoxically reports vasoconstriction.

The explanation for this paradoxical binarisation result lies in the concomitant change in angiogram signal intensity. Following stimulation, the angiogram intensity decreased significantly for all five vessels ($p < 0.0005$), with reductions ranging from −14.1% to −19.9% (Figs. 8g,h; Fig. 1). Because the binarised diameter is derived from the number of pixels exceeding a fixed threshold, a reduction in signal intensity directly translates to fewer above-threshold pixels and hence a smaller apparent diameter, even if the true vessel diameter has increased.

To quantify this confounding, the correlation between diameter change and angiogram intensity change was computed for each method (Figs. 9, 1). For the binarised method, the per-vessel correlations were uniformly positive and strong (ρ = 0.59, 0.80, 0.27, 0.46, 0.52 for vessels 1–5), yielding an overall correlation of $\rho = 0.523$ ($p < 0.0005$). This indicates that the binarised diameter measurement is substantially confounded by the angiogram signal intensity. In contrast, the Gaussian model yielded an overall correlation of $\rho = -0.101$ and the GG model yielded $\rho = -0.098$, confirming that the model-based diameter measurements are effectively independent of angiogram signal intensity fluctuations. These results demonstrate that model-based fitting not only provides more accurate and repeatable diameter measurements, but also recovers physiologically meaningful vascular responses that binarisation-based methods can misrepresent.

Table 1. Summary of flicker stimulation results. Top: percentage diameter change per vessel for each method together with corresponding p-values. Bottom: Pearson correlation (ρ) between measured diameter and angiogram signal intensity for each method and vessel. Overall correlations: Gaussian = −0.101, GG = −0.098, BW = 0.523.

Change in Diameter	Vessel 1		Vessel 2		Vessel 3		Vessel 4		Vessel 5	
Gaussian (% Change)	**9.2**	$p < 0.0005$	**9.6**	$p = 0.0019$	1.5	$p = 0.59$	**28.5**	$p < 0.0005$	**4.9**	$p = 0.0008$
Gen. Gaussian (% Change)	1.5	$p = 0.28$	**9.9**	$p < 0.0005$	−0.6	$p = 0.79$	**28.4**	$p < 0.0005$	**3.1**	$p = 0.0034$
Binarised (% Change)	**−12.5**	$p < 0.0005$	**−10.2**	$p < 0.0005$	**−4.2**	$p = 0.0011$	1.0	$p = 0.404$	**−9.3**	$p < 0.0005$
Angiogram Intensity (% Change)	**−14.1**	$p < 0.0005$	**−18.2**	$p < 0.0005$	**−14.3**	$p < 0.0005$	**−19.9**	$p < 0.0005$	**−16.3**	$p < 0.0005$

Correlation to Intensity	Vessel 1		Vessel 2		Vessel 3		Vessel 4		Vessel 5	
Gaussian Diameter Corr.	−0.07	$p = 0.48$	**0.23**	$p = 0.019$	**−0.28**	$p = 0.0044$	−0.15	$p = 0.14$	**−0.24**	$p = 0.018$
Gen. Gaussian Diameter Corr.	0.03	$p = 0.79$	0.07	$p = 0.47$	**−0.28**	$p = 0.0056$	−0.14	$p = 0.16$	−0.18	$p = 0.081$
Binarised Diameter Corr.	**0.59**	$p < 0.0005$	**0.80**	$p < 0.0005$	**0.27**	$p = 0.0077$	**0.46**	$p < 0.0005$	**0.52**	$p < 0.0005$

4. Discussion

4.1. Maximum Intensity or Mean Intensity?

4.2. Future Directions

A future direction of this work to to move toward fully automated segmentation and calculation vessel diameters from enface angiogram maps. While ROIs were manually selected in this work, it is also possible to automatically segment angiographic data using traditional skeletonization methods, which are commonly used for automated OCTA analysis. From a skeletonized image, vessel angle can be computed for further extraction of vascular cross-sections perpendicular to the vessel axis. These cross-sections could then be processed using the model-based methods described here to enable fully automated analysis of 3D OCTA data sets, rather than the semi-automated approach shown here for 2D OCTA data. This type of model-based approach is particularly important for comparing 3D volumes acquired at different time points in the mouse eye as there are often intensity losses caused by drying of the eye, cataract formation, etc. As we have shown in this work, even low levels of intensity loss occurring over a short time window can significantly impact the measured vessel diameter when using a binarization approach, but not when using the models presented here.

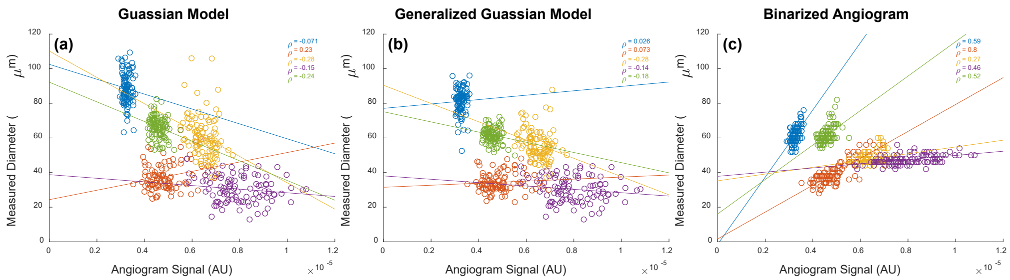

Fig. 9. Measured vessel diameter as a function of angiogram signal intensity during functional flicker stimulation for (a) Gaussian model, (b) generalised Gaussian model, and (c) binarised angiogram. Each colour represents one of five vessels; ρ denotes the Pearson correlation coefficient. The binarised method shows strong positive correlations between diameter and signal intensity, indicating confounding, whereas model-based methods show weak or no correlation.

4.3. Computation Time

It is clear from the results that model-based approaches outperform traditional diameter measurements based on a binarized image. Nevertheless, the improved linearity and repeatability come at the cost of computation time. Fitting a model to the raw aniogram data is significantly more computationally expensive and may not always be the most appropriate choice if results of the processing are time sensitive. Nevertheless, these approaches are highly parallelizable given the independent nature of the measurements, so it is expected that compuation time could be significantly accelerated with appropriate hardware.

4.4. Threshholding

The results presented here used a basic, global threshold for the binarization of the angiogram data. There are numerous other approaches for binarizing the angiogram, which will most likely each behave differently. For example, angiogram signals can be enhanced with vesselness filters and dynamic thresholds can be used to binarize images with varying SNR. When applying a vesselness filter, it is possible to enhance the shape of the angiogram and omit noise, but it is also possible for such enhancement to overcompensate the signal, leading to larger than expected vessel appearance. Dynamic thresholding can also yield a more uniform angiogram appearance, but similarly risks over- or undercompensating vessels given that threshold level directly affects the measured diameter. Further analysis of these other methods would be required to determine their efficacy in returning accurate and precise diameter values.

4.5. Literature Comparison

Discuss the paper [11]: ‘Maximum value projection produces better enface OCT angiograms than mean value projection’.

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